

Schrödinger Bridge for Non-linear ROMs

Results on the Bump and the NACA0012, and the Warm-Start Pay-off

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Recap — Fokker–Planck Formulation

Let $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ be the initial and terminal marginals, time rescaled to $[0, 1]$, $\gamma > 0$ the diffusion coefficient.

We seek a density–velocity pair $(\rho_t, \mathbf{b}_t)_{t \in [0, 1]}$ with $\rho_0 = \mu$, $\rho_1 = \nu$ such that

- ρ_t solves the **Fokker–Planck equation**

$$\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{b}_t) = \gamma \Delta \rho_t,$$

- the pair minimizes the **expected kinetic energy**

$$\mathcal{J}(\rho, \mathbf{b}) = \int_0^1 \int_{\mathbb{R}^d} \frac{1}{2} |\mathbf{b}_t(\mathbf{x})|^2 \rho_t(\mathbf{x}) \, d\mathbf{x} \, dt.$$

Girsanov applied to $dX_t = \mathbf{b}_t(X_t) \, dt + \sqrt{2\gamma} \, dW_t$ gives $\text{KL}(\mathbb{P} \parallel \mathbb{R}) = \mathcal{J}/(2\gamma)$: the Schrödinger bridge is the minimal-energy stochastic interpolation between μ and ν .

Recap — Heat and Advection–Diffusion Semigroups

Hopf–Cole ($\eta = e^f$, $\eta^* = e^g$) linearizes the optimality system into two transport–diffusion PDEs, whose transition operators are the SB kernel.

Driftless bridge ($\beta \equiv 0$) — heat semigroup $\mathcal{P}_\tau = e^{\tau\gamma\Delta}$

$$\partial_\tau \phi = \gamma \Delta \phi, \quad \partial_n \phi|_{\partial\Omega} = 0 \quad (\text{Neumann, no-flux}).$$

Drifted bridge — advection–diffusion semigroups $\mathcal{Q}_\tau = e^{\tau\mathcal{L}}$, $\mathcal{Q}_\tau^\dagger = e^{\tau\mathcal{L}^*}$

$$\underbrace{\partial_\tau \phi = \beta \cdot \nabla \phi + \gamma \Delta \phi}_{\mathcal{Q}: \text{ non-conservative, backward}}, \quad \underbrace{\partial_\tau \phi = -\nabla \cdot (\beta \phi) + \gamma \Delta \phi}_{\mathcal{Q}^\dagger: \text{ conservative, forward}}.$$

IPFP in log-potential form:

$$f^{(k+1)} = \log \mu - \log \mathcal{Q}_1[e^{g^{(k)}}], \quad g^{(k+1)} = \log \nu - \log \mathcal{Q}_1^\dagger[e^{f^{(k+1)}}].$$

FVM: TPFA diffusion + first-order upwind advection, SSP-RK2, Anderson-accelerated.

$$\beta \equiv 0 \Rightarrow \mathcal{Q}_1 = \mathcal{Q}_1^\dagger = \mathcal{P}_1.$$

Recap — Convex Displacement Interpolation (CDI)

The maps act on the shock-feature marginals; the physical Mach field is recovered gradient-free. For $t \in (0, 1)$,

$$\hat{M}(t, \mathbf{x}) = (1 - t) M_0((1 - t) \mathbf{x} + t S(\mathbf{x})) + t M_1(t \mathbf{x} + (1 - t) T(\mathbf{x}))$$

- M_0, M_1 = endpoint Mach fields, sampled at the displaced pre-images by linear interpolation on the mesh;
- exact at the endpoints: $\hat{M}(0, \cdot) = M_0$, $\hat{M}(1, \cdot) = M_1$;
- no map inversion — forward pre-image $t\mathbf{x} + (1 - t)T$ pulls back M_1 , backward pre-image $(1 - t)\mathbf{x} + tS$ pulls back M_0 , blended with weights $(1 - t, t)$.

The reference drift is designed to keep T, S close to the identity, so the CDI reconstruction stays well-conditioned at small γ .

Recap — Construction of the Transport Maps T, S

T is the conditional mean of the target position given the source (built from g through Q_1); S is its backward analogue (from f through $Q_1^\dagger$):

$$T(\mathbf{x}) = \frac{Q_1[\text{id } e^g](\mathbf{x})}{Q_1[e^g](\mathbf{x})} - \beta_{\text{bias}}(\mathbf{x}), \quad S(\mathbf{x}) = \frac{Q_1^\dagger[\text{id } e^f](\mathbf{x})}{Q_1^\dagger[e^f](\mathbf{x})} - \beta_{\text{bias}}(\mathbf{x}).$$

Neumann identity-leak correction

$$\beta_{\text{bias}} = \frac{Q_1[\text{id}]}{Q_1[1]} - \Phi_{\mathbf{b}},$$

measured against the drift's own *reflected deterministic flow* $\Phi_{\mathbf{b}}$ (identity when $\mathbf{b} \equiv 0$), so $\mathbf{b}$'s transport stays in T, S .

Evaluated in the stable log-domain with a positivity shift $\text{id} \mapsto \text{id} - c$; a trust gate reverts kernel-collapse outlier cells to the identity. $T, S \rightarrow \text{id}$ far from the transported feature.

Oblique Drift (1/2) — Wave Angles

The LE and TE shocks rotate between their M_0 and M_1 wave angles:

θ - β - M relation (weak root) — LE family

$$\tan \theta = 2 \cot \beta_s \frac{M^2 \sin^2 \beta_s - 1}{M^2(\gamma_g + \cos 2\beta_s) + 2}, \quad \beta_s^{\text{LE}}(M) = \beta_s(\alpha, M).$$

TE family — LE shock $\rightarrow$ Prandtl-Meyer(2α) $\rightarrow$ TE shock

$$\nu(M_b) = \nu(M_a) + 2\alpha, \quad \beta_s^{\text{TE}}(M) = \beta_s(\alpha, M_b(M)).$$

Ray angles and rotation rates, $M(t) = (1 - t)M_0 + tM_1$

$$\psi_{\text{LE}} = \beta_s^{\text{LE}}, \quad \psi_{\text{TE}} = \beta_s^{\text{TE}} - \alpha, \quad \omega_\kappa(t) = \left. \frac{d\beta_s^\kappa}{dM} \right|_{M(t)} (M_1 - M_0).$$

Oblique Drift (2/2) — Windowed Rigid Rotation

Four windowed rigid rotations about the apexes $\mathbf{a}_\kappa$

$$\boldsymbol{\beta}(t, \mathbf{x}) = \sum_{\kappa \in \{\text{LE, TE}\}} \sum_{\sigma = \pm 1} \sigma \omega_\kappa(t) w_{\kappa, \sigma}(t, \mathbf{x}) \mathbf{R}(\mathbf{x} - \mathbf{a}_\kappa), \quad \mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$w_{\kappa, \sigma} = e^{-\text{dist}_{\kappa, \sigma}^2 / \sigma_w(s)^2} \mathbf{1}[s_{\kappa, \sigma} > 0], \quad s_{\kappa, \sigma} = (\mathbf{x} - \mathbf{a}_\kappa) \cdot \hat{\mathbf{d}}_{\kappa, \sigma}.$$

Zero in the freestream; the conical band $\sigma_w(s)$ covers the swept wedge; $\nabla \cdot \boldsymbol{\beta} \neq 0 \Rightarrow$ enters through the conservative $Q^\dagger$.

What Is New Since the γ -Sweep

The diamond established the method; the question was whether it survives a geometry it was not designed around.

Bump (channel)

- Wall-mounted circular arc, 14,178 cells, $h = 0.025$.
- **No incidence axis** — AoA is meaningless on a channel wall.
- **No oblique drift** — it is the analytic θ - β - M wedge solution, and there is no wedge.
- 22 Mach bands, $0.80 \rightarrow 3.00$, width 0.10.

NACA0012 (airfoil)

- Rounded leading edge, 23,572 cells, $h = 0.025$.
- **Detached bow shock** at every Mach — the first geometry where the transported feature is curved and never attaches.
- Two incidences: AoA 0° and 2° .
- 22 Mach bands per incidence.

Everything else is held fixed: `iso_hessian/eig` marginal, γ -ladder down to 10^{-4} , drifts `{null, ffd, SBsquared_exact+ffd}`, and the same three interpolators compared in every run — **BaryCDI** (SB), **Linear**, **FFD**.

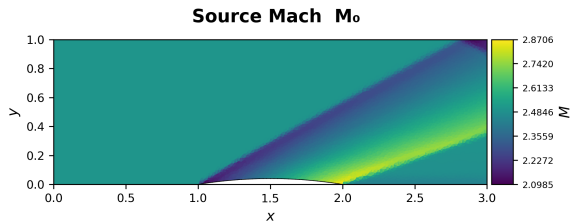
- Each run interpolates one 0.10-wide Mach band from its two endpoints and is scored against **9 high-resolution Euler solves** at $t = 0.1, \dots, 0.9$.
- Metrics per run: L^2 , L^∞ , W_2 , C_D , C_L , shock-band L^2 , shock angle $\Delta\beta$, IPFP residual, and the cost split (offline vs. online).
- **FFD is the control that matters**: it is the registration the drift is built from, used directly as an interpolator. If SB does not beat FFD, the bridge is not paying for its IPFP cost.

Three physical regimes

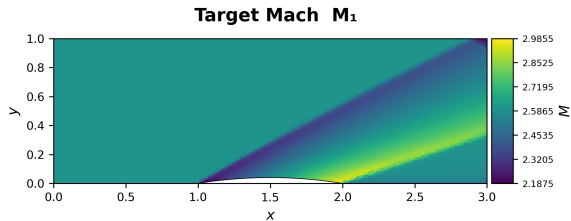
$M < 1.0$	no shock at all
1.0–1.25	detached bow shock
$M > 1.25$	attached oblique shock

The transport premise (a marginal concentrated on a moving shock) is built for the third. The first two are *exploratory* and are shaded red/orange in every sweep plot — not dropped, so the failure mode stays visible.

Bump — the Two Marginals, $M : 2.50 \rightarrow 2.60$

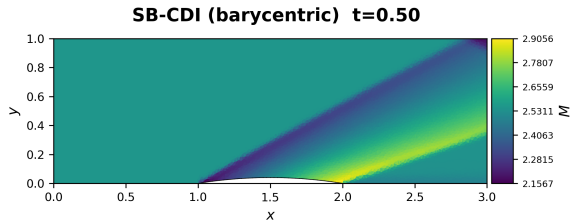


Endpoint $M_0 = 2.50$

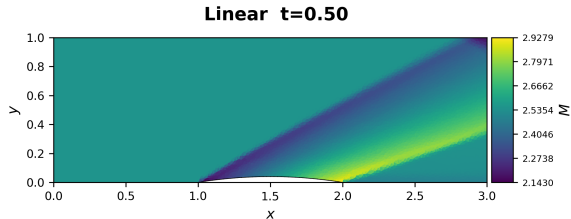


Endpoint $M_1 = 2.60$

Bump — Reconstruction at $t = 0.5$ ($M_\infty = 2.55$)

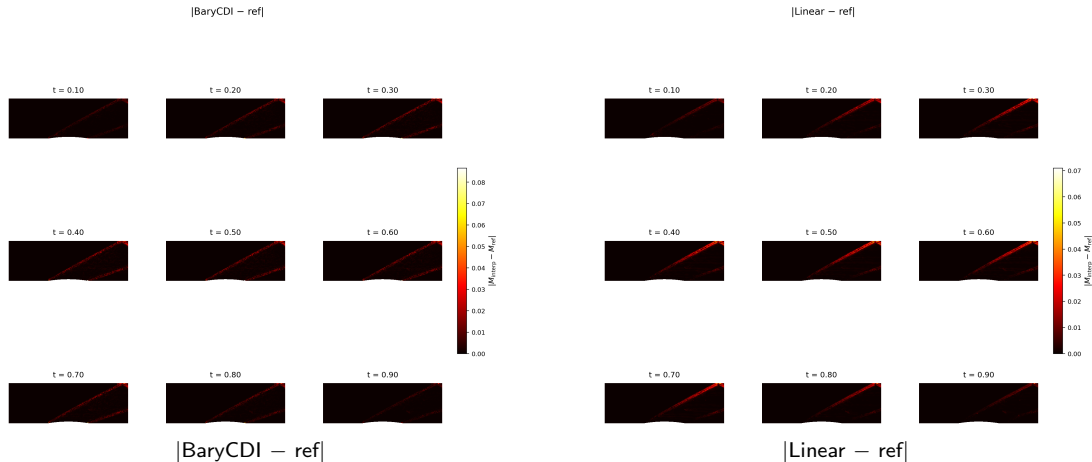


BaryCDI (SB) — the shock sits on the reference ray



Linear — the two endpoint shocks survive as a smeared pair

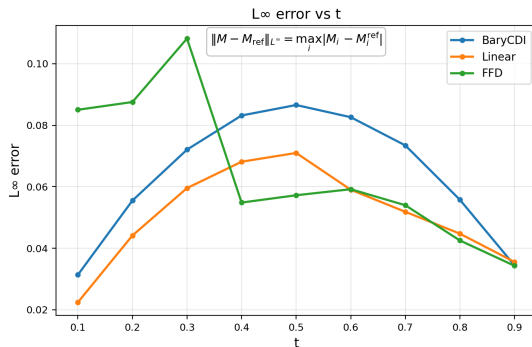
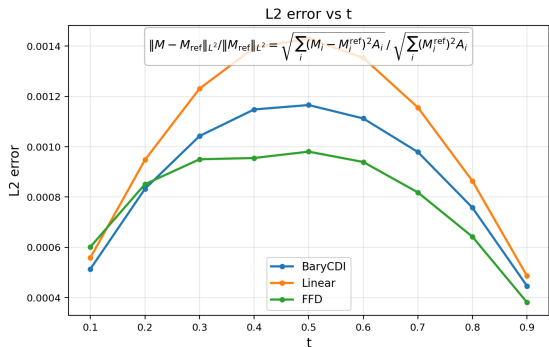
Bump — Absolute Error vs. the Reference



Both leave error *on* the shock — that is the residual grid alignment. The difference is the area between the two shock positions, which Linear fills and SB does not.

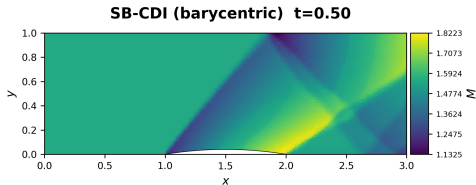
Bump — Error vs. t , $M : 2.50 \rightarrow 2.60$

Mach interpolation error vs. high-resolution reference

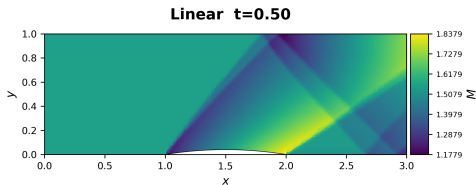


L^2 : SB is below Linear across the middle of the interval, $1.18\times$ on the band mean, and the whole error budget is down at 10^{-3} — this is the regime where the interpolation is genuinely accurate, not just relatively better. L^∞ : **Linear wins** ($7.1e-2$ vs. $8.7e-2$); the peak error is the shock cell itself, and moving the shock does not make that cell thinner.

Bump — a Second Band, $M : 1.50 \rightarrow 1.60$ ($M_\infty = 1.55$ at $t = 0.5$)



BaryCDI (SB)

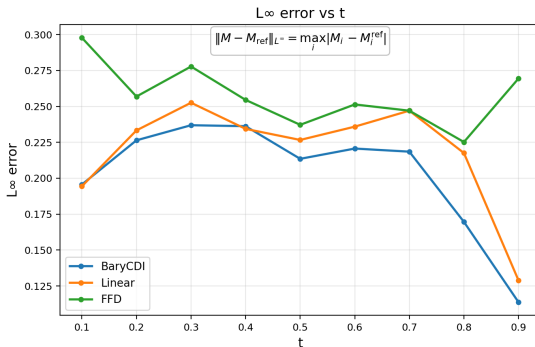
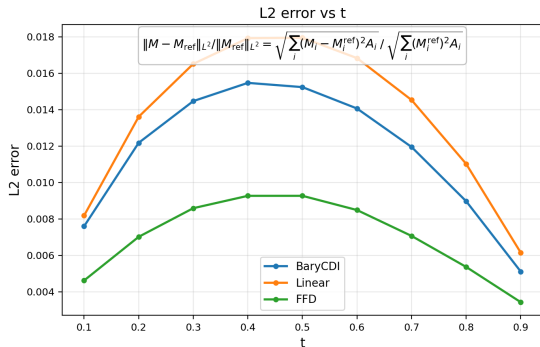


Linear

At $M = 1.5$ the reflected wave is still inside the domain, so the field carries *two* interacting shock systems instead of one — a harder transport than the $M = 2.55$ band, and the drift-free bridge still resolves both.

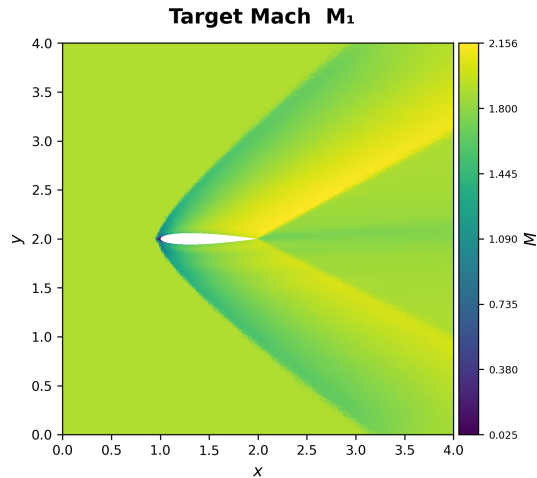
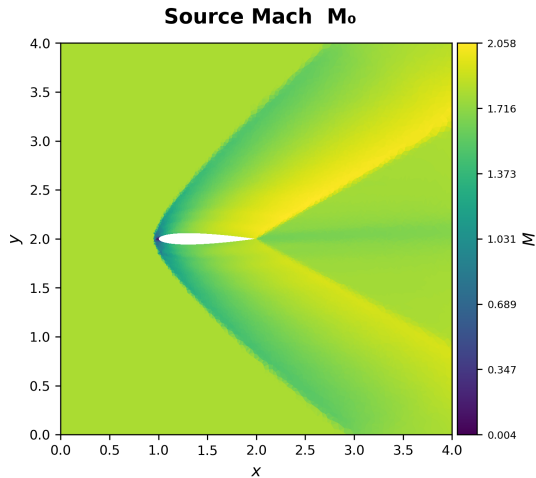
Bump — Error vs. t , $M : 1.50 \rightarrow 1.60$

Mach interpolation error vs. high-resolution reference

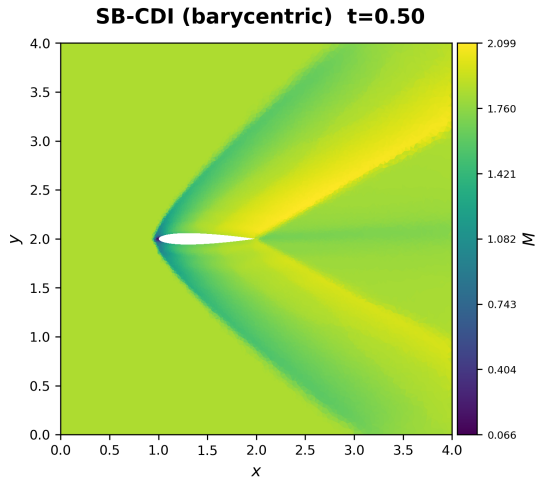


L^2 $1.17\times$ below Linear, and here SB also takes $\max_t L^\infty$ (0.237 vs. Linear 0.253 and FFD 0.298) — the opposite of the $M : 2.50 \rightarrow 2.60$ band. All three drifts converge on this band (res $\approx 9\text{e-}8$), so this is the bridge at its budget, not at its cap.

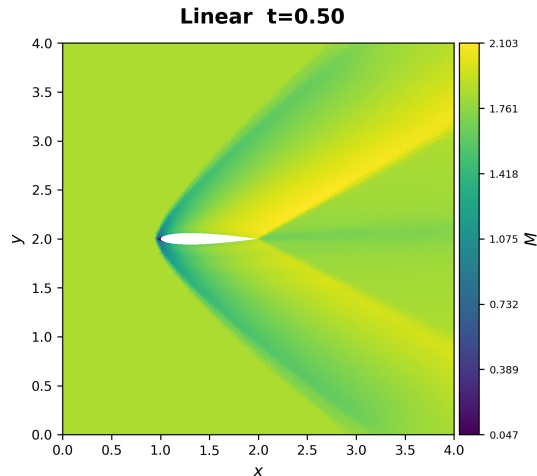
NACA0012 — the Two Marginals, AoA 2°, $M : 1.80 \rightarrow 1.90$



NACA0012 — Reconstruction at $t = 0.5$ ($M_\infty = 1.85$, AoA 2°)



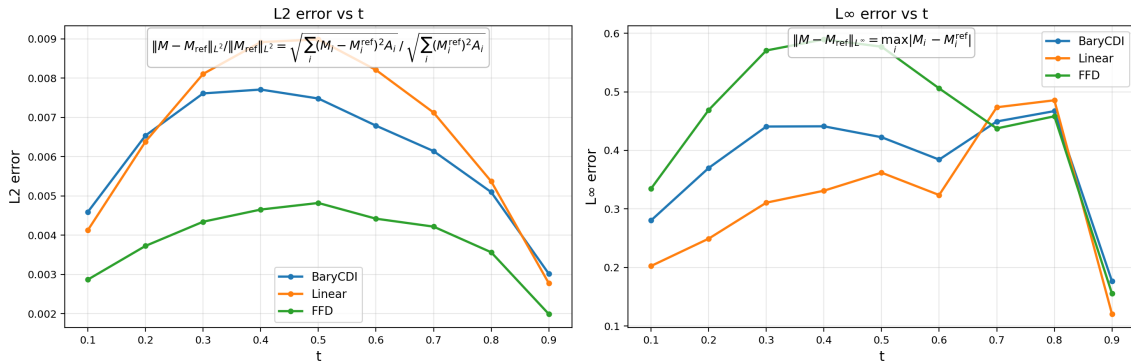
BaryCDI (SB)



Linear

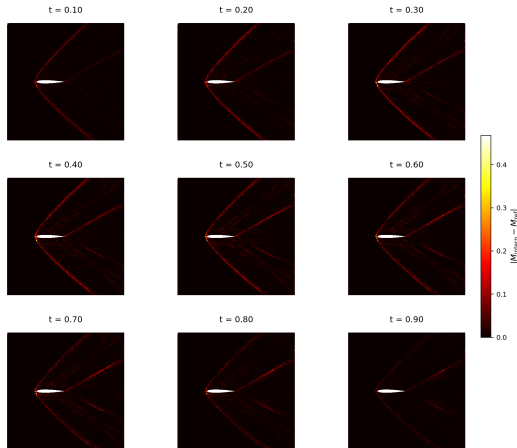
NACA0012 — Error vs. t , AoA 2° , $M : 1.80 \rightarrow 1.90$

Mach interpolation error vs. high-resolution reference



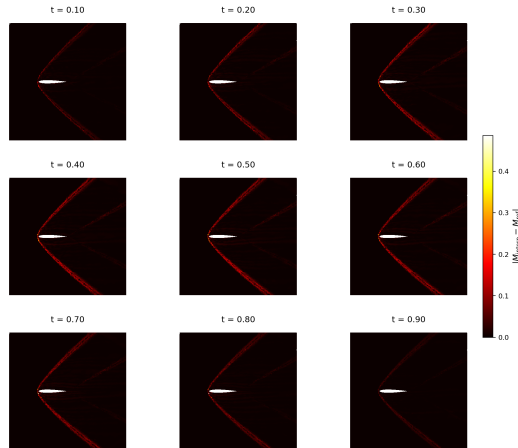
NACA0012 — Absolute Error, AoA 2° , $M : 1.80 \rightarrow 1.90$

|BaryCDI — ref|



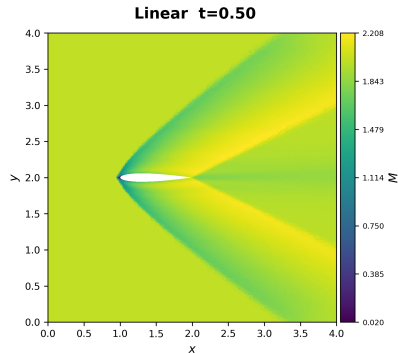
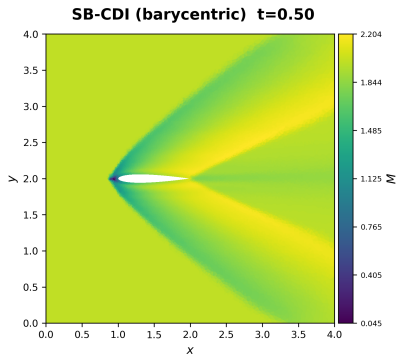
|BaryCDI — ref|

|Linear — ref|



|Linear — ref|

NACA0012 — the Other Axis: AoA $0^\circ \rightarrow 1^\circ$ at $M = 2.00$

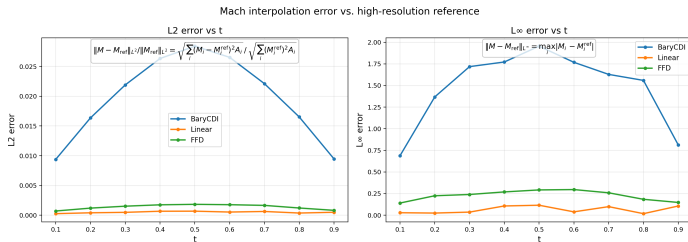


BaryCDI (SB) at $t = 0.5$ — AoA 0.5°

Linear at $t = 0.5$

Same machinery, same marginal, same code path — only the interpolation *axis* changes: the two endpoints now differ by 1° of incidence instead of 0.10 in Mach. Drift null, $\gamma_{\min} = 10^{-2}$ (the only AoA band of this case run to completion).

NACA0012 — the AoA Axis Is Where the Method Breaks



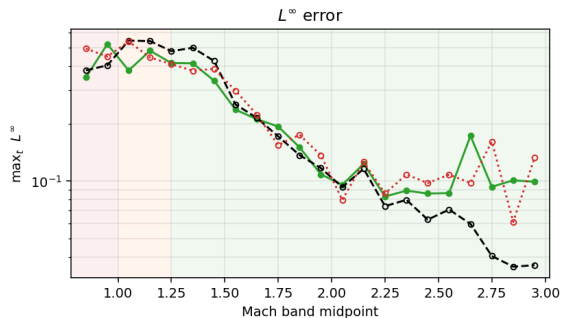
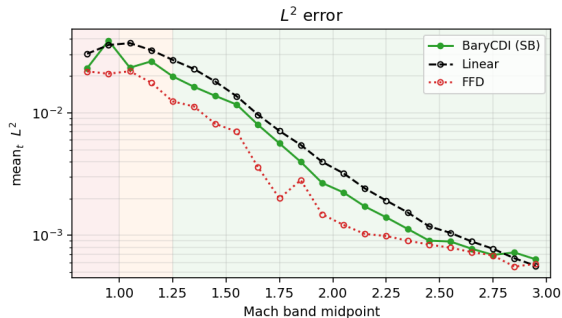
SB is 40 $\times$ worse than the linear blend

	BaryCDI	Linear	FFD
mean L^2	1.97e-2	4.98e-4	1.38e-3
$\max_t L^\infty$	1.95	0.115	0.296

A 1 $^\circ$ change in incidence barely *moves* the wave system — it changes its **amplitude**, and rotates the body itself. The Hessian marginal sees almost no displacement to transport, so the bridge invents one and drags the wall with it. **Transport is the wrong prior for this axis.**

Bump — L^2 and L^∞ across the Whole Mach Sweep

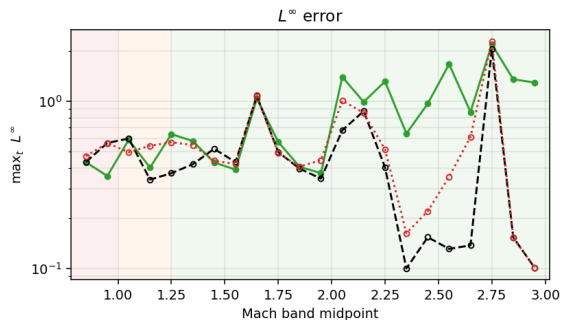
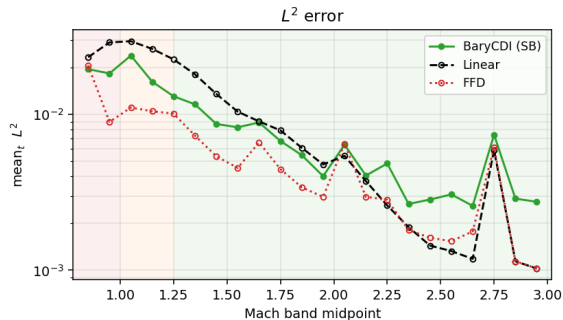
bump — Mach-interval sweep · $h=0.025$ · $\gamma_{\min}=10^{-4}$ · AoA 0° · drift = null
red: no shock orange: detached bow shock green: attached oblique shock



L^2 : SB is below Linear in **19 of 22** bands, median gain $1.31\times$, and the ordering is stable across the whole supersonic range. L^∞ : a genuine tie (SB ahead in 9/22) — the peak error is one shock cell, and transporting the shock does not shrink it. **FFD stays below SB at every Mach.**

NACA0012 — L^2 and L^∞ , AoA 0°

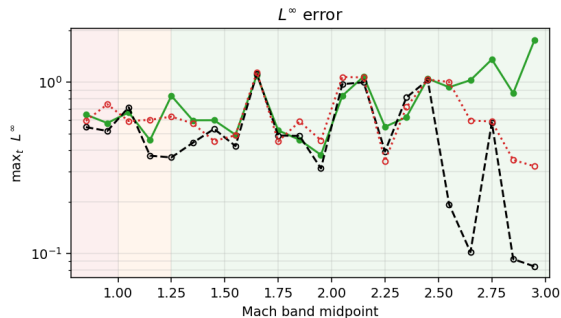
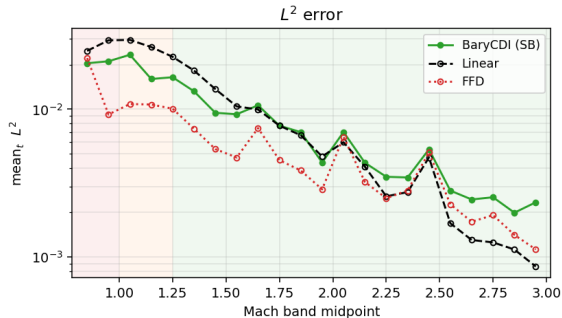
naca0012 — Mach-interval sweep · $h=0.025$ · $\gamma_{\min}=10^{-4}$ · AoA 0° · drift = SBsquared_exact+ffd
red: no shock orange: detached bow shock green: attached oblique shock



The airfoil splits in two at $M \approx 2$: below it SB wins on L^2 (up to $1.73\times$), above it Linear wins. The L^∞ panel says why — past $M \approx 2$ SB's peak error jumps to $\mathcal{O}(1)$ while Linear's does not.

NACA0012 — L^2 and L^∞ , AoA 2°

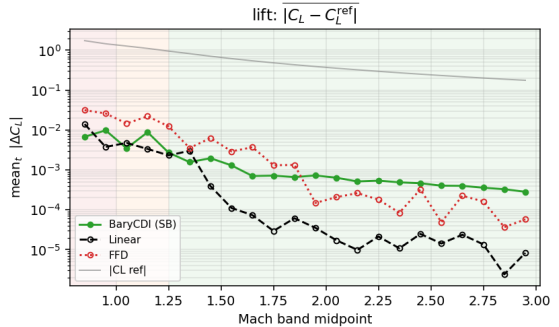
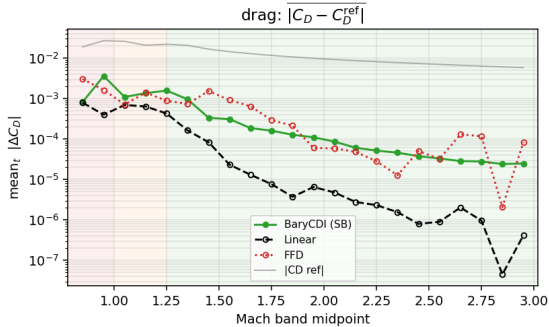
naca0012 — Mach-interval sweep · $h=0.025$ · $\gamma_{\min}=10^{-4}$ · AoA 2° · drift = null
red: no shock orange: detached bow shock green: attached oblique shock



Same crossover at incidence, same location. The L^∞ spikes line up with the Mach values where the reference C_D itself is known to jump: at $h = 0.025$ the bow-shock standoff spans only 1.3–5 cells, so the captured shock snaps between cell rows. **That is a mesh artefact in the reference, and SB inherits it amplified.**

Bump — Aerodynamic Coefficients across the Sweep

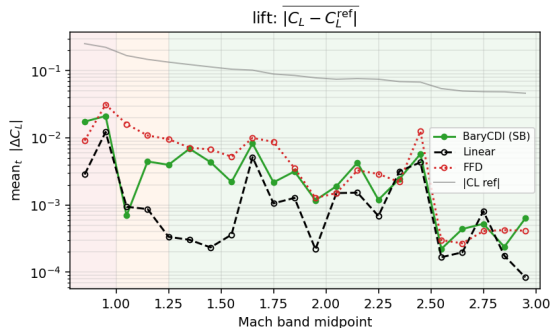
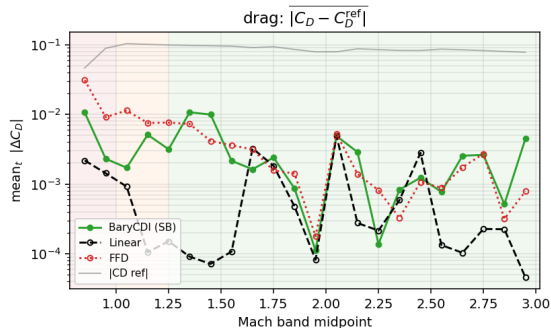
bump — aerodynamic coefficients across the Mach sweep · $\gamma_{\min} = 10^{-4}$ · AoA 0° · drift = null



Grey = the reference level the errors are read against. In the validated band ($M > 1.25$): $|\Delta C_D|$ is **1.47%** of C_D^{ref} for SB, 0.21% for Linear, 2.29% for FFD. **Linear wins the integrated forces by roughly 5×**, even in the bands where it loses the field by 1.4×

NACA0012 — Aerodynamic Coefficients, AoA 2°

naca0012 — aerodynamic coefficients across the Mach sweep · $\gamma_{\min} = 10^{-4}$ · AoA 2° · drift = null



Same verdict on the airfoil: $\overline{|\Delta C_D|} = 3.22\%$ (SB) vs. 1.02% (Linear) vs. 2.72% (FFD) of C_D^{ref} .

The Sweep in One Table

Means over the **validated** bands ($M > 1.25$, 18 bands), $\gamma_{\min} = 10^{-4}$, best drift per case.

case	drift	mean L^2			$ \overline{\Delta C_D} / C_D^{\text{ref}}$	
		BaryCDI	Linear	FFD	BaryCDI	Linear
bump	null	5.16e-3	6.77e-3	3.17e-3	1.47%	0.21%
naca, AoA 0°	SB ² +ffd	5.91e-3	6.57e-3	4.01e-3	6.67%	0.96%
naca, AoA 2°	null	6.33e-3	6.70e-3	4.16e-3	3.22%	1.02%

Where SB wins

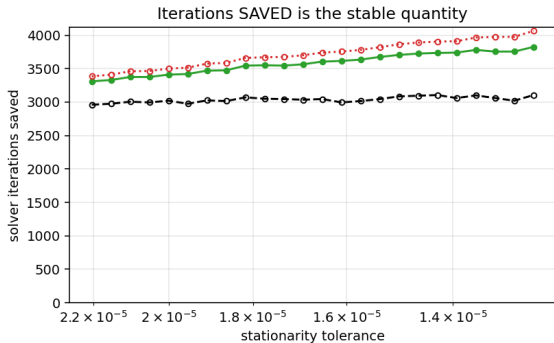
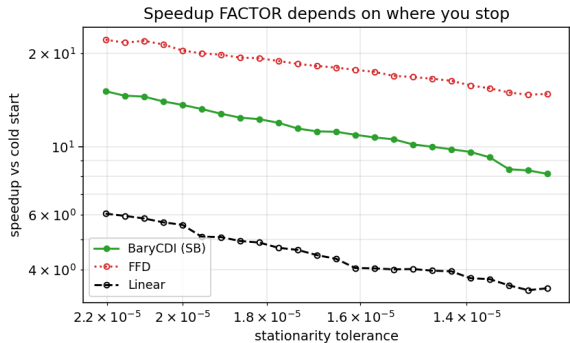
- L^2 vs. Linear on the bump: **19/22** bands.
- L^2 vs. Linear on the airfoil for $M < 2$: up to **1.73×**.
- Shock *position* — the physical quantity the marginal encodes.

Where SB loses

- vs. **FFD**, at every Mach, on both geometries.
- L^∞ : tie on the bump, clear loss on the airfoil above $M \approx 2$.
- C_D , C_L : Linear by 3–7×

Speed-Up Is a Property of the Tolerance, Not of the Seed

Warm-start benefit · diamond $h=0.025$ · M 2.00→2.50 · $t=0.5$ · seeds from the ffd run



Speed-Up Recap — the Numbers

Diamond $h = 0.025$, $M : 2.00 \rightarrow 2.50$, averaged over the 9 interpolation times. Cold baseline in iterations; 0.889 ms per solver step.

seed	loosest reachable tol $2.2\text{e-}5$		tightest reachable tol $1.27\text{e-}5$	
	iterations	speed-up	iterations	speed-up
cold (baseline)	3648	$1.0\times$	4536	$1.0\times$
Linear	354	$101.5\times$	1057	$5.9\times$
BaryCDI (SB)	158	$29.7\times$	483	$10.6\times$
FFD	118	$36.0\times$	294	$15.7\times$

Stable across the band

seed	saved (min)	saved (max)
BaryCDI	3240	4775
FFD	3300	4910
Linear	2915	4535

The *factor* spans $10\times$ for SB and $231\times$ for Linear across the reachable band; the *iterations saved* spans $1.5\times$ for all three. Report saved iterations, with the floor stated.

Pricing It: When Does the Bridge Pay for Itself?

One SB run (3-stage ladder to $\gamma = 10^{-4}$, all stages converged) costs 21.9 s offline and 0.23 s to emit all 9 fields. At the tightest reachable tolerance:

seed	offline (s)	solve/query (s)	saved/query (s)	break-even vs. cold
cold	—	4.03	—	—
Linear	0.00	0.94	3.09	immediate
BaryCDI (SB)	21.88	0.43	3.60	6 queries
FFD	0.005	0.26	3.77	immediate

The comparison that matters is not against cold

- Against **cold**, SB repays its IPFP after **6 queries** — it works.
- Against **Linear**, which is free, SB buys only 0.51 s per query, so the 21.9 s ladder repays after **~43 queries**.
- Against **FFD**, which costs 5 ms offline and saves *more*, SB **never** repays.

The warm-start benefit is real and large, but it is a benefit of *having a good initial guess*, and the cheapest good guess wins.

Summary

- 1 The method **transfers** to two geometries it was not built for: a wall-mounted bump with no wedge and no incidence axis, and a rounded-nose airfoil with a permanently detached bow shock.
- 2 On the **field** in L^2 it does what it claims — SB beats the linear blend in **19/22** bump bands and up to **1.73×** on the airfoil below $M \approx 2$ — by placing the shock rather than fading between two of them.
- 3 On **integrated forces** it does not: Linear is $3\text{--}7\times$ better on C_D , because the marginal carries no mass where the forces are computed.
- 4 As a **warm start** the interpolated field is worth ~ 3600 solver iterations, a $10.6\times$ speed-up at the tightest reachable tolerance, repaying the IPFP after 6 queries against a cold solve.
- 5 The open problem is unchanged and now confirmed on three geometries: **SB does not beat the FFD registration it is conditioned on**, at any γ , on any mesh.